Jeffrey Yang

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Education

University of British Columbia
Bachelor of Science in Computer Science — Sept. 2023 – Present
- **4.10/4.33 GPA**; Trek Excellence Scholarship Award

Richmond Secondary School
International Baccalaureate Diploma Programme — Sept. 2021 – June 2023

Experience

Research Assistant — May 2026 – Present
Cho Lab & NERVES Lab, ICORD — Remote
- **Automated markerless rodent gait analysis** for spinal cord injury research, connecting video calibration, DeepLabCut tracking, knee correction, and gait measurements in a PySide6 application.
- Extended MyoAssist's MuJoCo model from **22 to 28 muscles**, adding six lumbar muscles so the simulated torso can bend forward and backward; implemented tendon paths, muscle controls, and checks for joint motion and torque.
- Improved **terrain-transition prediction from 46 sensor channels** for adaptive switching, using temporal-difference learning to identify patterns in level walking, stairs, and turns; measure how early predictions detect stair and level-ground transitions across participants.

WordPress Website Designer — April 2025 – Present
VeryShop — Remote
- Customized PHP plugin logic and HTML/CSS templates for a WooCommerce wholesale storefront with 10 location-based product collections, adapting product displays and purchasing behaviour and refining layouts across affected pages.
- Tested **retailer-only product access and ordering**, checking server-side authentication guards that restrict the full catalogue and wholesale pricing to registered accounts; verified page and plugin changes against those access requirements.

Thrift Store Clerk — Sept 2021 – May 2023
Regional Animal Protection Society — Richmond, BC
- Handled customer purchases, cash-register transactions, donations, merchandise stock, and store display.

Projects

DLC Gait Assembler | ***Python, PySide6, DeepLabCut*** — 2026
- Built a desktop tool that extracts **up to 178 gait parameters** from DeepLabCut coordinates across left, right, and bottom camera views: 132 hindlimb features plus 46 optional forelimb/interlimb parameters covering stride timing, limb movement, and coordination.
- Applied confidence filtering, gap interpolation, and Butterworth smoothing before stride analysis; exported per-stride measurements, session summaries, and SVG figures, with previews for checking tracking and calibration.
- Added PCA and Random Forest analysis with cross-validation grouped by animal, generating feature-importance rankings and confusion matrices while ensuring each animal appears only in training or test data within a fold.

Brainwave Surfing — EEG Platformer | ***Python, JavaScript, HTML/CSS*** — Nov. 2025
- Built a browser game that **turns recorded EEG into playable terrain**, with sleep and seizure recordings controlling wave height, player collisions, and audio.
- Wrote a JavaScript EDF parser that reads binary recordings, selects EEG channels, converts samples to physical units, and resamples signals to 50 Hz for gameplay; used Python/MNE to convert recordings and sleep-stage annotations into JSON.

WooDisplay | ***Swift, SwiftUI, Core Graphics*** — 2026
- Built a macOS app that **converts WooCommerce exports into printable product catalogues**; validated CSV columns and merged product variants with their parent products to preserve price ranges, stock status, and attributes, with brand, category, price, and stock filters.
- Used shared pagination for the live preview and Core Graphics PDF export, including category-specific pages and product ordering; fetched product images concurrently and saved reusable layouts and colours as JSON themes.

DS ROM Music Arrangement Patcher | ***Java, Swing, Binary Patching*** — Sept. 2025
- Built a Java/Swing editor for a **236-entry trainer-music table** in Pokemon Black 2 and White 2, letting users choose which battle theme plays for each trainer class.
- Automated ROM extraction, decompression, and rebuilding; detected game versions and applied an existing music-assignment patch with user-selected ARM9 table edits and overlay-size corrections.

See 8 more projects

Technical Skills

Languages: TypeScript, Java, Python, Swift, C, C++, R, SQL, JavaScript, HTML/CSS, PHP
Frameworks/Libraries: SwiftUI, PySide6, DeepLabCut, MuJoCo, MNE, NumPy, pandas, scikit-learn, Matplotlib, Seaborn, Node.js, Swing
Tools: WordPress, WooCommerce, Conda, CrystalTile2, Claude Code, Codex, Cursor, GitHub Copilot, Langflow, Aider
Extras: REST APIs, JSON, English, Mandarin, French